TEJAS TOVINKERE PATTABHI

Staff Engineer · Cloud · Distributed Systems · Applied AI

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Field of endeavor: Data Processing and Analysis Techniques; Advanced Modeling and Simulation.

SUMMARY

Staff Engineer with 10+ years turning large-scale data and compute problems into reliable production systems. At Qualcomm, works on the critical path of the autonomous-driving program: the infrastructure and applied AI that validate the software running in production vehicles. Named co-inventor on two granted U.S. patents.

CORE SKILLS

Languages	Python · Java · C# · C · Bash	**Cloud**	AWS · Google Cloud (Dataflow) · Azure
Orchestration	Kubernetes · Argo Workflows/Events · Docker · AWS CDK	**Applied AI**	LLM agents (Claude Agent SDK · Bedrock) · agentic coding (Claude Code · Codex) · custom Claude skills/plugins
ML & vectors	Self-hosted embeddings (Qwen3) · pgvector · semantic dedup		

EXPERIENCE

Qualcomm

Sep 2021 – Present

Staff Engineer · San Diego, CA

Build the large-scale infrastructure that replays recorded drive data to validate automated-driving software: perception, radar, vehicle self-motion/positioning, ground-truth labeling, and automatic emergency braking.

- **Automotive-grade reprocessing clusters.** Designed, built, and operate two on-prem Kubernetes clusters of Qualcomm **QCR-100** cards (the same Snapdragon Ride silicon that ships in vehicles) for bit-accurate ADAS/AD reprocessing, running **~100–300× faster than general-purpose compute**, fast enough to replay recorded drives at close to real time.
- **Reprocessing-as-a-Service platform.** Runs identically in cloud or on-prem: Kubernetes job orchestration, DAG dependency resolution, event-driven **Argo Workflows**, parallel petabyte-scale data transfer, and automated failure classification. Also integrated the QCR-100 into custom hosted cloud environments for on-demand access.
- **Key collaborator, Qualcomm–BMW Group program.** Helped deliver the reprocessing pipeline behind the jointly developed Snapdragon Ride Pilot system (publicly launched Sept. 2025, debuting in the BMW iX3).
- **Production AI triage agent** (project lead), tens of thousands of failures/day: self-hosted embeddings + **pgvector** semantic dedup, then a **Claude Agent SDK + AWS Bedrock** agent that diagnoses root cause and opens Jira tickets / GitHub fix-PRs, escalating ambiguous cases to humans.
- **Fleet-health & throughput monitoring.** Prometheus stack tracking each accelerator's health/firmware and **Real-Time Factor (RTF)**, processing time relative to the recorded drive's duration; plus IPMI/BMC power control and safe node lifecycle.

OpenText (formerly Webroot)

Sep 2015 – Sep 2021

Senior Software Engineer · San Diego, CA

Led internet-scale cybersecurity and threat-intelligence systems.

- Led distributed clusters that safely crawled and threat-classified a **peak of 17M+ URLs/day**; co-invented "DeepCrawl," which discovers related malicious resources from a known link and classifies threat level with an ML model: **basis of two granted U.S. patents**.
- Built an IP threat-classification system and a cloud-reputation scoring system, and architected multi-cloud honeypots (AWS, GCP, Azure) capturing live attacks to feed both. Owned AWS hosting/DevOps.

Earlier: Award Solutions: UT Dallas OpenStack project, open-sourced (2015) · TradeStation Technologies: SDE Intern, AWS EMR log analytics (2014) · Société Générale: Windows Server Administrator, Bangalore (2012–2013).

PATENTS · PUBLICATIONS · PEER REVIEW · EDUCATION

U.S. Patents 12,452,259 B2 (2025) **& 11,201,875 B2** (2021): "*Web Threat Investigation Using Advanced Web Crawling,*" named co-inventor, Open Text Inc. · First-author publication, *IJITCS*, 2013 (spatial-clustering algorithm) · Reviewer, IEEE GSCon 2026 (two papers, on invitation).

Ph.D. Artificial Intelligence, Univ. of the Cumberlands · in progress (started 2026) · **M.S. Computer Science**, UT Dallas · 2015 · **B.E. Information Science**, PES / VTU, India · 2012 · First Class with Distinction (all 8 semesters)